Protocol Suites

3.3.1

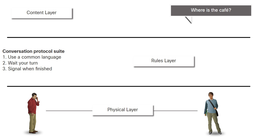
Network Protocol Suites

In many cases, protocols must be able to work with other protocols so that your online experience gives you everything you need for network communications. Protocol suites are designed to work with each other seamlessly.

A protocol suite is a group of inter-related protocols necessary to perform a communication function.

One of the best ways to visualize how the protocols within a suite interact is to view the interaction as a stack. A protocol stack shows how the individual protocols within a suite are implemented. The protocols are viewed in terms of layers, with each higher-level service depending on the functionality defined by the protocols shown in the lower levels. The lower layers of the stack are concerned with moving data over the network and providing services to the upper layers, which are focused on the content of the message being sent.

As illustrated in the figure, we can use layers to describe the activity occurring in face-to-face communication. At the bottom is the physical layer where we have two people with voices saying words out loud. In the middle is the rules layer that stipulates the requirements of communication including that a common language must be chosen. At the top is the content layer and this is where the content of the communication is actually spoken.



Protocol suites are sets of rules that work together to help solve a problem.

3.3.2

Evolution of Protocol Suites

A protocol suite is a set of protocols that work together to provide comprehensive network communication services. Since the 1970s there have been several different protocol suites, some developed by a standards organization and others developed by various vendors.

During the evolution of network communications and the internet there were several competing protocol suites, as shown in the figure.

TCP/IP Layer Name	TCP/IP	ISO	AppleTalk	Novell Netware
Application	HTTP DNS DHCP FTP	ACSE ROSE TRSE SESE	AFP	NDS
Transport	TCP UDP	TP0 TP1 TP2 TP3 TP4	ATP AEP NBP RTMP	SPX
Internet	IPv4 IPv6 ICMPv4 ICMPv6	CONP/CMNS CLNP/CLNS	AARP	IPX
Network Access	Ethernet ARP WLAN			

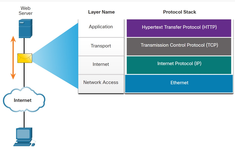
- **Internet Protocol Suite or TCP/IP** – This is the most common and relevant protocol suite used today. The TCP/IP protocol suite is an open standard protocol suite maintained by a standards organization (IETF and IAB).
- **Open Systems Interconnection (OSI) protocols** – This is a family of protocols developed jointly in 1977 by the International Organization for Standardization (ISO) and the International Telecommunications Union (ITU). The OSI protocol also included a seven-layer model called the OSI reference model. The OSI reference model categorizes the functions of its protocols. Today OSI is mainly known for its layered model. The OSI protocols have largely been replaced by TCP/IP.
- **AppleTalk** – A short-lived proprietary protocol suite released by Apple Inc. in 1985 for Apple devices. In 1995, Apple adopted TCP/IP to replace AppleTalk.
- **Novell NetWare** – A short-lived proprietary protocol suite and network operating system developed by Novell Inc. in 1983 using the IPX network protocol. In 1995, Novell adopted TCP/IP to replace IPX.

3.3.3

TCP/IP Protocol Example

TCP/IP protocols are available for the application, transport, and internet layers. There are no TCP/IP protocols in the network access layer. The most common network access layer LAN protocols are Ethernet and WLAN (wireless LAN) protocols. Network access layer protocols are responsible for delivering the IP packet over the physical medium.

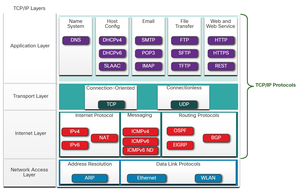
The figure shows an example of the three TCP/IP protocols used to send packets between the web browser of a host and the web server. HTTP, TCP, and IP are the TCP/IP protocols used. At the network access layer, Ethernet is used in the example. However, this could also be a wireless standard such as WLAN or cellular service.



3.3.4

TCP/IP Protocol Suite

Today, the TCP/IP protocol suite includes many protocols and continues to evolve to support new services. Some of the more popular ones are shown in the figure.



TCP/IP is the protocol suite used by the Internet and the networks of today. TCP/IP has two important aspects for vendors and manufacturers:

- **Open standard protocol suite** – This means it is freely available to the public and can be used by any vendor on their hardware or in their software.
- **Standards-based protocol suite** – This means it has been endorsed by the networking industry and approved by a standards organization. This ensures that products from different manufacturers can interoperate successfully.

Click each button for a brief description of protocols at each layer.

Application Layer | Transport layer | Internet Layer | Network Access Layer

Network Access Layer

Address Resolution

- **ARP** – Address Resolution Protocol. Provides dynamic address mapping between an IPv4 address and a hardware address.

Note: You may see other documentation state that ARP operates at the Internet Layer (OSI Layer 3). However, in this course we state that ARP operates at the Network Access layer (OSI Layer 2) because it's primary purpose is the discover the MAC address of the destination. A MAC address is a Layer 2 address.

Data Link Protocols

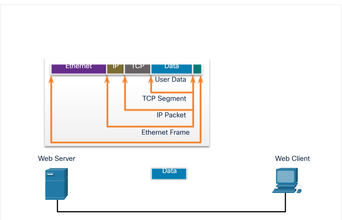
- **Ethernet** – Defines the rules for wiring and signaling standards of the network access layer.
- **WLAN** – Wireless Local Area Network. Defines the rules for wireless signaling across the 2.4 GHz and 5 GHz radio frequencies.

3.3.5

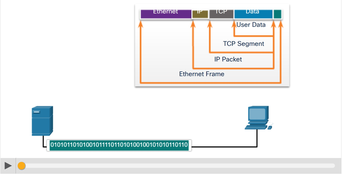
TCP/IP Communication Process

The animations in the figures demonstrate the complete communication process using an example of a web server transmitting data to a client.

Click the Play in the figure to view an animation of a web server encapsulating and sending a web page to a client.



Click the Play in the next figure to view an animation of the client receiving, and de-encapsulating the web page for display in the web browser.



3.3.6

Check Your Understanding – Protocol Suites

Check your understanding of protocol suites by choosing the BEST answer to the following questions.

1. UDP and TCP belong to which layer of the TCP/IP protocol?

- ( ) application
- (•) transport
- ( ) internet
- ( ) network access

2. Which two protocols belong in the TCP/IP model application layer?

- [ ] EIGRP
- [x] DNS
- [ ] OSPF
- [ ] ICMP
- [x] DHCP

3. Which protocol operates at the network access layer of the TCP/IP model?

- ( ) HTTP
- ( ) IP
- ( ) DNS
- (•) Ethernet

4. Which of the following are protocols that provide feedback from the destination host to the source host regarding errors in packet delivery? (Choose two.)

- [ ] IPv4
- [ ] TCP
- [x] ICMPv4
- [ ] IPv6
- [ ] UDP
- [x] ICMPv4

5. A device receives a data link frame with data and processes and removes the Ethernet information. What information would be the next to be processed by the receiving device?

- ( ) HTTP at the application layer
- ( ) HTML at the application layer
- (•) IP at the internet layer
- ( ) UDP at the internet layer
- ( ) TCP at the transport layer

6. Which services are provided by the internet layer of the TCP/IP protocol suite? (Choose three.)

- [ ] File Transfer
- [ ] Address Resolution
- [x] Routing Protocols
- [x] Messaging
- [ ] Ethernet
- [x] Internet Protocol

Check

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